Date: 05/05/2026 | **MONTHLY CHALLENGE** | **SET A**

Duration: 2 hours | **III YEAR – SEMESTER 6**

Question 1:

1. Problem Title & Link

Title: Shortest Path with Limited Wall Breaks

2. Problem Statement

You are given a grid of size m x n:

- 0 → empty cell (you can walk)
- 1 → wall (you cannot walk unless you break it)

You start at the **top-left (0,0)** and want to reach **bottom-right (m-1,n-1)**.

You are allowed to **break at most k walls**.

• Find the **minimum number of steps** required to reach the destination.

• If it is not possible, return -1.

3. Examples (Input → Output)

Example 1

Input:

grid = [

[0,1,0],

[0,1,0],

[0,0,0]

]

k = 1

Output:

4

Explanation: Break one wall and reach destination in shortest path.

4. Constraints

- 1 <= m, n <= 40
- 0 <= k <= m*n
- Grid contains only 0 and 1
- Movement allowed: up, down, left, right
- Must find **shortest path**

Question 2:

1. Problem Title & Link

Title: Minimum Time to Execute Tasks with Cooldown & Priority

2. Problem Statement

You are given an array of tasks:

tasks = ["A","A","A","B","B","C"]

Each task:

- Takes **1 unit of time**
- Same tasks must have at least **k cooldown time** between them

You can:

- Execute one task per time unit
- OR stay idle

Each task also has a **priority weight**:

weights = { "A": 3, "B": 2, "C": 1 }

When multiple tasks are available, you must pick:

- **Highest frequency**
- If tie → pick **higher weight**

Return the **minimum time required** to complete all tasks.

3. Examples

Example 1

Input:

tasks = ["A","A","A","B","B","C"]

k = 2

weights = {"A":3,"B":2,"C":1}

Output:

8

4. Constraints

- 1 <= len(tasks) <= 10^5
- 0 <= k <= 10^4
- Tasks are uppercase characters
- Weights are positive integers
- Must be efficient → **O(n log n)** expected